

## 1 — Automatically Maintain Student Count

```
CREATE DATABASE CollegeDB;
USE CollegeDB;

CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Student_Name VARCHAR(50),
  Course VARCHAR(50)
);
CREATE TABLE Student_Count (
  Total_Students INT
);
INSERT INTO Student_Count VALUES (0);
DELIMITER //
CREATE TRIGGER after_student_insert
AFTER INSERT ON Student
FOR EACH ROW
BEGIN
  UPDATE Student_Count
  SET Total_Students = Total_Students + 1;
END //
DELIMITER ;
INSERT INTO Student VALUES
(101, 'Navya', 'BCA');
INSERT INTO Student VALUES
(102, 'Neha', 'BCA');
INSERT INTO Student VALUES
(103, 'Manya', 'BCA');

SELECT * FROM Student;

SELECT * FROM Student_Count;
```

|                |              |         |
|----------------|--------------|---------|
| Result Grid    | Filter Rows: | Export: |
| Total_Students |              |         |
| 0              |              |         |

## 2 — Maintain an Automatic Salary Audit

```
CREATE DATABASE CompanyDB;
USE CompanyDB;
```

```
CREATE TABLE Employee (
Emp_ID INT PRIMARY KEY,
Emp_Name VARCHAR(100),
Department VARCHAR(100),
Salary DECIMAL(10,2)
);
CREATE TABLE Salary_Audit (
Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
Emp_ID INT,
Old_Salary DECIMAL(10,2),
New_Salary DECIMAL(10,2),
Change_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO Employee VALUES
(1, 'Navya', 'IT', 50000);
INSERT INTO Employee VALUES
(2, 'Neha', 'HR', 45000);
DELIMITER //
CREATE TRIGGER after_salary_update

AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
IF OLD.Salary <> NEW.Salary THEN
INSERT INTO Salary_Audit
(Emp_ID, Old_Salary, New_Salary)
VALUES
(OLD.Emp_ID, OLD.Salary, NEW.Salary);
```

```

END IF;
END //
DELIMITER ;
UPDATE Employee
SET Salary = 60000
WHERE Emp_ID = 1;
UPDATE Employee
SET Salary = 50000
WHERE Emp_ID = 2;
SELECT * FROM Employee;
SELECT * FROM Salary_Audit;

```

| Result Grid                                                                                                                                                                                                                                                        |          |        |            |            |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|------------|------------|---------------------|
| Filter Rows: <input type="text"/>                                                                                                                                                                                                                                  |          |        |            |            |                     |
| Edit:    Export |          |        |            |            |                     |
|                                                                                                                                                                                                                                                                    | Audit_ID | Emp_ID | Old_Salary | New_Salary | Change_Time         |
| ▶                                                                                                                                                                                                                                                                  | 1        | 1      | 50000.00   | 60000.00   | 2026-09-29 12:17:53 |
|                                                                                                                                                                                                                                                                    | 2        | 2      | 45000.00   | 50000.00   | 2026-09-29 12:17:53 |
| ✱                                                                                                                                                                                                                                                                  | NULL     | NULL   | NULL       | NULL       | NULL                |

### 3 — Prevent Deletion of Important Employees

```

DROP DATABASE IF EXISTS CompanyDB2;
CREATE DATABASE CompanyDB2;
USE CompanyDB2;
CREATE TABLE Employee (
    Emp_ID INT PRIMARY KEY,
    Emp_Name VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(10,2)
);
CREATE TABLE Deleted_Employee_Log (
    Log_ID INT AUTO_INCREMENT PRIMARY KEY,
    Emp_ID INT,
    Emp_Name VARCHAR(50),
    Deleted_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO Employee VALUES (101, 'Navya', 'IT', 50000);
INSERT INTO Employee VALUES (102, 'Neha', 'HR', 45000);
INSERT INTO Employee VALUES (103, 'Manya', 'Management', 80000);
INSERT INTO Employee VALUES (104, 'Spoorthi', 'Finance', 60000);

```

```

DELIMITER //
CREATE TRIGGER before_employee_delete
BEFORE DELETE ON Employee
FOR EACH ROW
BEGIN
IF OLD.Department = 'Management' THEN
SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Management employees
cannot be deleted';
END IF;
END //
DELIMITER ;
DELIMITER //
CREATE TRIGGER after_employee_delete
AFTER DELETE ON Employee
FOR EACH ROW
BEGIN
INSERT INTO Deleted_Employee_Log (Emp_ID, Emp_Name) VALUES
(OLD.Emp_ID, OLD.Emp_Name);
END //
DELIMITER ;

DELETE FROM Employee WHERE Emp_ID = 101;
DELETE FROM Employee WHERE Emp_ID = 103;
SELECT * FROM Employee;
SELECT * FROM Deleted_Employee_Log;

```

| Result Grid  |        |        |          |                     |
|--------------|--------|--------|----------|---------------------|
| Filter Rows: |        |        |          |                     |
| Edit:        |        |        |          |                     |
|              | Log_ID | Emp_ID | Emp_Name | Deleted_Time        |
| ▶            | 1      | 101    | Navya    | 2026-09-29 12:22:46 |
| *            | NULL   | NULL   | NULL     | NULL                |